

Figure glossary — what every label on the 3×3 plots means

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Purpose

This note explains every label, code word, and visual element appearing on the 3×3 grids saved in `notebooks/analysis/`:

- `multimode_polarized_3x3.png`, `multimode_unpolarized_3x3.png` (and their `_convergence.png` companions)
- `constraint_pretrain_gmtkn55_demo_3x3.png` (the earlier pretraining-recipe × metric version)

It is intended as a stand-alone reference: open the figure on one screen and this file on another.

The big picture

We train a small **GGA-style exchange-correlation neural network** (call it the *nn-functional*) to reproduce reference exchange and correlation enhancement factors at every grid point of a precomputed PBE density. We then ask two questions about that nn-functional:

1. **How well does it reproduce real chemistry references** (BH76 reaction energies and W4-11 atomization energies from GMTKN55)?
2. **How much do physical-constraint priors and the pretraining target choice matter** — both for accuracy and for run-to-run reproducibility across random seeds?

The 3×3 plots answer those questions by sweeping *three independent axes* (self-consistency depth × constraint stack × pretraining-loss weighting) on *three independent metrics*. Each plot is one specific functional configuration (polarized or unpolarized correlation baseline).

Decoding the figure title

“Self-consistency ladder × constraints × pretraining vs GMTKN55 (polarized, 16 seeds)”

Phrase	Meaning
Self-consistency ladder	The <i>rows</i> (fixed- ρ , one-shot, 3-step SCF) — increasing the number of SCF cycles the nn-functional is run through before we read off the energy.

Phrase	Meaning
× constraints	The <i>x-axis</i> of each panel — a progression of physical-constraint priors switched on cumulatively (unconstrained → +L0 → +L0+UEG → +L0+UEG+NNc).
× pretraining	The <i>colored bars</i> inside each x-position — random initialization versus two flavors of pretrained network.
vs GMTKN55	The reference values for the left and right columns come from the GMTKN55 benchmark database (the Grimme group’s gold-standard collection of accurate post-HF energies).
polarized / unpolarized	Which version of the PW92 correlation baseline the nn-functional is built on. <i>Polarized</i> uses the spin-resolved ζ -dependent baseline (correct for open-shell atoms); <i>unpolarized</i> uses the $\zeta = 0$ total-density baseline. The two PNGs let us compare them directly.
16 seeds	The “random init” bars summarize 16 independent random network initializations (different seeds). The pretrained bars are a single pretrain seed (this is a demo, not a production benchmark).

Rows — the self-consistency ladder

The same trained network is evaluated three ways, in increasing self-consistency. ρ denotes the electron density; “ ρ_{PBE} ” is the precomputed PBE density.

Row label on the y-axis	Code term	What is actually evaluated
fixed-ρ	fixed_rho	The nn-functional is never solved self-consistently . We just evaluate its XC energy on the <i>frozen</i> PBE density. Zero Roothaan steps. The fastest, cheapest test; this is what the original constraint demo used.
one-shot (1 step)	one_shot / SolverMode.FIXED_J with max_cycles=1 and FeaturePolicy.FROZEN	Build the nn-functional’s Fock matrix from the PBE density once, with the Coulomb (J) term pinned to J[ρ_{PBE}] , then diagonalize once . The orbitals get to relax, but ρ does not feed back into J or the descriptor features.

Row label on the y-axis	Code term	What is actually evaluated
3-step SCF	3step / SolverMode.FULL with max_cycles=3 and FeaturePolicy.REASSEMBLE	A genuine 3-cycle self-consistent SCF: at every cycle the Coulomb term $J[\rho]$ <i>and</i> the nn-functional's descriptor features are rebuilt from the current density. This is “real” self-consistency, truncated to 3 cycles for cost.

Why a ladder. Reading the figure top-to-bottom tells you whether the conclusions you draw from the cheap fixed- ρ evaluation survive when you turn the SCF machinery back on. Empirically (see the report) 3-step $\approx$ fixed- ρ almost exactly on this set, so the cheap evaluation is faithful.

Equations — what each mode literally computes

Equations below have been cross-checked against the actual implementation (xcquinox/alec/oneshot.py, xcquinox/alec/solver.py, xcquinox/alec/solver_manual.py) — not just the docstring.

Notation. Let $\hat{h}_{\text{core}}$ be the one-electron core Hamiltonian (kinetic + nuclear-attraction), $J[D]_{\mu\nu} = \sum_{\lambda\sigma} (\mu\nu|\lambda\sigma) D_{\lambda\sigma}$ the Coulomb (Hartree) matrix built from an AO density matrix D , $V_{\text{xc}}^{\text{NN}}[\rho]$ the exchange-correlation potential matrix produced by the trained nn-functional, and $E_{\text{xc}}^{\text{NN}}[\rho] = \int \epsilon_{\text{xc}}^{\text{NN}}(\rho, |\nabla\rho|^2, \dots) d\mathbf{r}$ its energy functional. E_{nuc} is the nuclear repulsion. Subscripts PBE denote the precomputed PBE reference solution (frozen); subscripts n index Roothaan cycles. The linear-mixer parameter $\alpha = \frac{1}{2}$ (the SolverConfig default) throughout.

1. fixed- ρ — no SCF, energy-only evaluation on the frozen PBE density:

$$E_{\text{fixed-}\rho} = E_{\text{nuc}} + \text{Tr}[\hat{h}_{\text{core}} D_{\text{PBE}}] + \frac{1}{2} \text{Tr}[J[D_{\text{PBE}}] D_{\text{PBE}}] + E_{\text{xc}}^{\text{NN}}[\rho_{\text{PBE}}].$$

The nn-functional contributes only its *energy* — no Fock matrix is built, no diagonalization is run, the orbitals do not relax.

2. one-shot — SolverMode.FIXED_J + FeaturePolicy.FROZEN, exactly one Roothaan step. Build the Fock matrix once with J pinned to $J[D_{\text{PBE}}]$ and $V_{\text{xc}}^{\text{NN}}$ evaluated on the frozen PBE-density grid inputs:

$$F = \hat{h}_{\text{core}} + \underbrace{J[D_{\text{PBE}}]}_{\text{pinned}} + \underbrace{V_{\text{xc}}^{\text{NN}}[\rho_{\text{PBE}}]}_{\text{frozen } \rho, \text{ frozen descriptors}}, \quad FC = SC\varepsilon \implies D_{\text{new}},$$

then linear-mix once and read off the energy on the mixed density (with J still pinned):

$$D_{\text{mix}} = \frac{1}{2} D_{\text{new}} + \frac{1}{2} D_{\text{PBE}},$$

$$E_{\text{one-shot}} = E_{\text{nuc}} + \text{Tr}[\hat{h}_{\text{core}} D_{\text{mix}}] + \frac{1}{2} \text{Tr}[J[D_{\text{PBE}}] D_{\text{mix}}] + E_{\text{xc}}^{\text{NN}}[\rho(D_{\text{mix}})].$$

The orbitals are allowed to relax exactly once, but neither J nor the nn-functional's potential is rebuilt.

3. 3-step SCF — SolverMode.FULL + FeaturePolicy.REASSEMBLE, at most 3 Roothaan steps. Starting from $D_0 = D_{\text{PBE}}$, for $n = 0, 1, 2$:

$$F_n = \hat{h}_{\text{core}} + J[D_n] + V_{\text{xc}}^{\text{NN}}[\rho_n], \quad F_n C_{n+1} = S C_{n+1} \varepsilon_{n+1} \implies D_{n+1}^{\text{new}},$$

$$D_{n+1} = \frac{1}{2} D_{n+1}^{\text{new}} + \frac{1}{2} D_n, \quad \text{halt early if } |E_{n+1} - E_n| < 10^{-6} \text{ Ha.}$$

Each cycle recomputes the Coulomb matrix from the live density and reassembles the nn-functional’s descriptor features from the live density. The final energy is the standard Kohn-Sham expression at the converged (or cycle-3) density:

$$E_{\text{3-step}} = E_{\text{nuc}} + \text{Tr}[\hat{h}_{\text{core}} D_{\text{final}}] + \frac{1}{2} \text{Tr}[J[D_{\text{final}}] D_{\text{final}}] + E_{\text{xc}}^{\text{NN}}[\rho_{\text{final}}].$$

The 3-cycle cap is the unroll length passed to `jax.lax.scan` inside `run_manual_scf`; early-converged states are simply held constant for the remaining iterations.

Footnote — UKS and the descriptor list in this study. For spin-polarized (UKS) evaluation the Fock matrix is built per spin channel, $F^\sigma = \hat{h}_{\text{core}} + J[D^\alpha + D^\beta] + V_{\text{xc}}^{\text{NN},\sigma}[\rho^\alpha, \rho^\beta]$, with exchange spin-scaled per Oliver-Perdew and correlation taken from the spin-resolved (or $\zeta = 0$) baseline — this is the `split_exc_energy_uks` branch whose gate is exactly what the **polarized vs unpolarized** PNG pair toggles. For the figures in this glossary the descriptor list is empty (`precompute_fixed_density_data(spec, descriptors=(), ...)` in `multimode_constraint_eval.py:379`), so the distinction between FROZEN and REASSEMBLE in this run reduces to whether ρ and $|\nabla\rho|^2$ on the integration grid come from D_{PBE} (frozen) or from the live D_n (reassembled).

Code-word translation

- **Roothaan step** = one diagonalization of the Fock matrix (one SCF cycle).
- **Fock matrix** = the one-electron effective Hamiltonian whose eigenvectors are the molecular orbitals; for KS-DFT it contains the XC potential V_{xc} .
- **J** = the classical Coulomb (Hartree) matrix. “**Fixed-J** / **FIXED_J**” means J is computed once from ρ_{PBE} and never recomputed.
- **FROZEN / REASSEMBLE feature policy** = whether the nn-functional’s descriptor inputs (ρ , $|\nabla\rho|$, ...) are recomputed each SCF cycle (REASSEMBLE) or held fixed at their PBE-density values (FROZEN). FROZEN must accompany FIXED_J; REASSEMBLE must accompany FULL.

Columns — the three metrics

Each column is a different MAE (mean absolute error), in **kcal/mol**, against a different reference set. Lower bars = better.

Left column: BH76 reaction-energy MAE vs GMTKN55-BH76RC

- **BH76** = a 76-reaction benchmark of thermochemical reaction barriers, curated by Goerigk & Grimme. **BH76RC** is its *reaction-energy channel* (ΔE for products – reactants), as opposed to BH76’s standard *barrier heights*.
- We use **6 of the 76** BH76RC reactions here — a deliberately held-out “probe-C” subset whose three companion reactions are used in training. The 6 shown are therefore a **transfer test**, not a training metric.
- Reference values are the GMTKN55 W2-F12 / CCSD(T) reaction energies.

- **PBE = 8.1 kcal/mol** (dashed horizontal line); this is the standard PBE functional’s error on these 6 reactions and is our “what a respectable GGA looks like” yardstick.

Middle column: per-species $|E_{nn} - E_{PBE}|$ MAE

- For every single species in the pool (29 atoms + small molecules), compute the absolute difference between the *nn-functional’s* total energy and *PBE’s* total energy, then mean over species.
- This is a **deviation-from-PBE diagnostic, not an accuracy benchmark**. No ground-truth reference goes into it. It answers: *how close to the reasonable-looking PBE total-energy surface does the nn-functional land?*
- Why it is informative: reaction-energy MAEs cancel any systematic per-species XC error (Σ coefficients = 0 for a balanced reaction). The per-species metric refuses that cancellation and exposes how *individually* good or bad each total energy is — which is where constraints actually act.
- There is **no PBE baseline line** on this panel by construction (PBE deviates from itself by 0).

Right column: atomization-energy MAE vs GMTKN55 W4-11

- **W4-11** = a 140-molecule subset of GMTKN55 with very high-accuracy CCSD(T)-derived atomization energies (the W4 protocol, ~sub-kcal/mol).
- We use a **10-molecule closed-shell subset** (H_2 , H_2O , CH_4 , NH_3 , CO , N_2 , CO_2 , HF , C_2H_2 , C_2H_4). Each atomization energy is computed as $E(\text{molecule}) - \Sigma E(\text{constituent atoms})$, and the MAE is taken against the W4-11 reference value (kcal/mol).
- **PBE $\approx$ 10.45 kcal/mol** (dashed line) on this subset.
- This is the metric that most aggressively exposes **open-shell-atom physics** — the atoms (H, C, N, O, F) enter with no cancelling partner on the other side, so any error in their spin-resolved correlation shows up at full weight. That is why polarized-vs-unpolarized matters so much here.

x-axis (within each panel) — the constraint ladder

These labels are the cumulative stack of *physical-constraint priors* baked into the nn-functional’s architecture. Each level adds one more constraint to the previous; constraints are mathematical guarantees, not soft losses.

x-axis label	Code term	Physical meaning
unconstrained	lob_lim=None, empty constraint list	No physics priors imposed: the network’s outputs are free real numbers. (The built-in Lieb–Oxford “squash” is also disabled — otherwise even “no constraints” would already be LO-bounded.)

x-axis label	Code term	Physical meaning
+LO	lieb_oxford on the exchange net	Enforces the Lieb-Oxford bound — a rigorous lower bound on the exchange-correlation energy ($E_{xc} \geq -C \cdot \int \rho^{\{4/3\}}$). Implemented as a smooth output squash that prevents the network from emitting unphysically negative enhancement factors.
+LO+UEG	lieb_oxford + ueg_limit on the exchange net	Adds the uniform-electron-gas limit : in the slowly-varying (uniform) density limit, the exchange enhancement factor $F_x \rightarrow 1$ (i.e. the functional reduces to LDA). This is a Levy-Perdew-Sahni-style consistency condition.
+LO+UEG+NNc	adds non_negative_correlation on the <i>correlation</i> net	NNc = non-negative correlation enhancement — guarantees the correlation enhancement factor stays non-negative, ruling out unphysically positive correlation energies. (“NNc” reads as “Non-Negative correlation”.)

Letter suffixes you may see in older labels

- The “(x)” or “(c)” suffix in some legends (+LO(x), +LO+UEG+NNc(c)) marks which *network* the constraint acts on: **x = exchange net** (xnet), **c = correlation net** (cnet). They are the two halves of the nn-functional and are constrained separately.

Quick reading

- **+LO** = “make exchange not blow up.”
- **+LO+UEG** = “...and make it reduce to LDA in the uniform limit.”
- **+LO+UEG+NNc** = “...and make correlation not be positive.” This is the fully-constrained level.

Bars and error indicators (the legend)

Inside each panel you see three colored bar series at every x-tick plus several error indicators:

Visual	Code label	What it represents
Red bar (dark red, leftmost)	random init (mean)	Arithmetic <i>mean</i> of the metric over 16 independently random-initialized networks (same architecture/constraints, different seeds, no pretraining). Tells you what an untrained network of this constraint level looks like on average.
Faint gray cap at the top of the red bar	worst of seeds	The <i>worst</i> (largest, i.e. worst-MAE) value across the 16 seeds. Drawn as an upward-only error cap because we only care how bad the tail gets. Pulls back as constraints add.
Bold dark-red whisker centered on the red bar	$\pm$ std (seeds)	One standard deviation across the 16 seeds. Shrinks as constraints add $\rightarrow$ constraints buy <i>robustness</i> , even when they don't move the mean.
Blue bar (slightly to the right)	pretrained [unweighted]	A <i>single</i> , pretrained network, where the pretraining loss is a plain unweighted mean of squared residuals between the network and the PBE/LDA target enhancement factors at every grid point. (One seed, hence no error bar.)
Teal bar (rightmost)	pretrained [integration]	Same pretrained network architecture, but the pretraining loss is integration-weighted — each grid-point residual is multiplied by the DFT integration weight $\times$ density factors, so the loss approximates the <i>integrated</i> exchange/correlation-energy error, not a uniform point-by-point error.
Black dashed line	PBE (8.1) (or PBE (10.45))	The PBE functional's MAE on this metric. Bars below the line beat PBE. The PBE line is omitted in the deviation-from-PBE column.

Loss-weighting code translation

- **unweighted** — $L = (1/N) \sum_i (f_{nn}(x_i) - f_{target}(x_i))^2$ over grid points i . Treats every point equally.
- **integration** — $L = \sum_i w_i \cdot (f_{nn}(x_i) - f_{target}(x_i))^2$, with $w_i \propto$ DFT integration weight $\times$ density-weighting that converts the per-point residual into an approximate energy-integral residual. Tilts the optimizer toward getting *energy-relevant* regions right rather than tail regions where ρ is tiny.

The convergence figure (*_convergence.png)

These are the two-panel companion figures `multimode_polarized_convergence.png` and `multimode_unpolarized_convergence.png`. They diagnose **pretraining optimization difficulty**, separately for exchange and correlation.

Element	Meaning
Left panel title: pretraining steps-to-converge [unweighted]	Same metric as below, but with the <i>unweighted</i> pretraining loss.
Right panel title: pretraining steps-to-converge [integration]	...with the <i>integration-weighted</i> loss.
Red bar (xnet)	Number of gradient steps the exchange network took to first reach 1.05 $\times$ its eventual minimum loss (i.e. “within 5% of the best loss it ever achieves”). Lower = pretraining converged faster.
Blue bar (cnet)	Same, for the correlation network .
y-axis: steps to reach 1.05 $\times$ min loss	Out of 1000 total pretraining steps. A bar at ≈ 1000 means the network never really converged within budget.
x-axis	The same unconstrained $\rightarrow$ +LO+UEG+NNc constraint ladder.

This figure answers: *do constraints help or hurt the pretraining optimizer?* For the polarized config the picture is non-monotonic and interesting (LO accelerates exchange dramatically; NNc accelerates correlation dramatically). For the unpolarized config most levels hit the budget — a landscape difference between the two functional families.

The other 3 $\times$ 3 — constraint_pretrain_gmtkn55_demo_3x3.png

This older figure is **structured differently**. It is the *fixed- ρ -only* version, but with the rows reinterpreted as **pretraining recipe**, not self-consistency depth:

Row	Meaning
150-step pretrain (unpolarized)	Short pretraining run, unpolarized correlation baseline.
1000-step pretrain (unpolarized)	Longer pretraining, same unpolarized baseline.
1000-step pretrain + spin-polarized PW92c	Longer pretraining, switched to the polarized correlation baseline . This is the row that drops atomization MAE from $\sim 27 \rightarrow \sim 10$ kcal/mol — the “polarized baseline is what fixes atomization” punchline.

The **columns**, **x-axis**, **bar/error semantics**, and **PBE dashed lines** are identical to the multimode 3×3 . Only the row meaning changes.

Software / data anchors (for cross-reference)

Term you may see	Where it lives
GMTKN55-BH76RC	Benchmark file <code>bh76rc.json</code> , GMTKN55 distribution.
GMTKN55 W4-11	Benchmark file <code>w411.json</code> (atomization-energy subset).
PBE	Perdew–Burke–Ernzerhof GGA, the reference DFT functional throughout.
PW92 / PW92c	Perdew–Wang 1992 LDA correlation. “PW92c” emphasizes the <i>correlation</i> part; the spin-polarized form is the ζ -dependent VWN/PW92 expression.
GGA	Generalized-gradient-approximation rung of Jacob’s ladder — the functional class our nn lives in (uses ρ and $\nabla\rho$ but no kinetic-energy density).
<code>xcquinox.alec.solver</code>	Where <code>SolverConfig</code> , <code>SolverMode</code> , <code>FeaturePolicy</code> , and <code>SolverBackend</code> are defined.
<code>xcquinox.alec.pretrain</code>	Defines <code>PretrainSpec</code> , <code>loss_weighting</code> , integration weights.
<code>make_constraint_levels()</code>	In <code>constraint_pretrain_gmtkn55_demo.py</code> — produces the four constraint-ladder labels.
<code>make_multimode_figure.py</code>	The plotting script that <i>literally produces</i> the multimode 3×3 and convergence PNGs from the saved JSON; this glossary is faithful to its label strings.

Advisor summary — what to take away from the multimode 3×3 figures

Across all three self-consistency rungs (fixed- $\rho \rightarrow$ one-shot $\rightarrow$ 3-step SCF) the picture is essentially identical — running the functional self-consistently barely moves any bar, so the cheap fixed- ρ conclusions hold. Within each rung, layering on physical constraints does not change the random-init *mean* accuracy but cuts seed-to-seed standard deviation by roughly 4× and the worst-of-16-seeds tail by roughly 2×, while pretraining (especially the integration-weighted variant) is what actually closes the absolute gap to PBE on BH76 and W4-11. Comparing the two PNGs, the spin-polarized correlation baseline is the binding factor for atomization energies: the polarized functional lands on PBE (≈ 10 kcal/mol) at the fully-constrained pretrained level, while the unpolarized one is stuck at ~ 27 kcal/mol and actually *worsens* as constraints are added — because open-shell atoms need ζ -resolved correlation the unpolarized functional cannot represent.